



MoF: A Background-Aware Multi-source Fusion Financial Trend Forecasting Mechanism

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Abstract

With the rapid growth of economic globalization and digital economics, accurately predicting stock price fluctuations has become crucial yet challenging due to high volatility and market noise. Existing forecasting methods, relying primarily on time-series data, technical indicators, and sentiment analysis, often fail to capture the semantic depth of background knowledge, particularly the influence of real-time events. To address this limitation, we propose MoF, a background-aware multi-source fusion mechanism for financial trend forecasting. MoF integrates stock price data with background knowledge on the impact of real-time events on stock trends, which includes key information from policy documents and stock commentaries, and subsequently leverages the MacBERT model to generate feature vectors for stock prediction. Our results show that MoF, by incorporating the influence of real-time events, improves accuracy and interpretability in stock trend forecasting, surpassing LSTM-based models with over 90% accuracy in predicting market fluctuations and providing reliable directional predictions.

Keywords Background-aware · LLM · Macbert · Financial trend

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1 Introduction

With the rapid development of economic globalization and the rise of the digital economy, stock prices, as key indicators of economic growth and market conditions, have become increasingly important in academic and investment circles. Accurate forecasting is crucial for investors, financial analysts, and policymakers, but stock prices exhibit high volatility and noise and influenced by factors, e.g., national exchange rates, fiscal policies, market sentiment, and macroeconomic conditions. Consequently, accurately forecasting stock prices present a significant challenge.

Research methods for predicting stock prices include traditional statistical models, machine learning techniques, deep learning approaches, and sentiment analysis methods. Traditional statistical models primarily include the autoregressive integrated moving average (ARIMA) (Box et al., 1976) and the generalized autoregressive conditional heteroskedasticity (GARCH) (Bollerslev, 1986) models. However, these traditional statistical models are inadequate for fitting non-linear, high-dimensional data and are prone to noisy data, resulting in significant prediction errors when forecasting real-time, dynamic stock prices.

Machine learning and deep learning methods have been extensively utilized in stock price prediction research in recent years. Machine learning techniques provide advantages in handling large datasets, capturing complex nonlinear relationships, and automatically learning feature representations. Researchers have employed traditional machine learning algorithms, e.g., logistic regression (LR) (Wu et al., 2021), gradient boosting decision tree (GBDT) (Zhou et al., 2019), support vector machine (SVM) (Gang, 2019), and neural networks. Deep learning models possess powerful feature learning and pattern recognition capabilities, can automatically learn complex feature representations from large-scale data, and offer higher flexibility and scalability. Applications of these models in stock price prediction include Recurrent Neural Networks (RNN) (Elman, 1990), Long Short-Term Memory Networks (LSTM) (Hochreiter & Schmidhuber, 1997), Convolutional Neural Networks (CNN) (LeCun et al., 1998), Deep Neural Networks (DNN), and Transformers (Vaswani et al., 2017).

These machine learning and deep learning algorithms primarily rely on historical data, with models trained to recognize patterns and regularities to forecast future price movements. However, these approaches have limitations as they neglect external textual factors such as social news, policy changes, or market sentiment. Stock price movements are closely linked to such factors. For instance, the completion of the merger between the North and South Railways in China sparked optimistic expectations for the railway industry, driving up stock prices. In contrast, an incident involving tanker trucks loaded with cooking oil caused a sharp drop in the stock price of Yihai Kerry Golden Dragon Fish Food Group Co, which hit a record low in a short period. The impact of such singular events or policy changes on the market is abrupt and non-cyclical, making it difficult to capture their effects using historical data alone. Existing stock price prediction methods primarily rely on historical stock price data, technical indicators,

and sentiment analysis, using machine learning and deep learning techniques. Although these methods have had some success, relying solely on traditional machine learning techniques to analyze sentiment indicators overlooks the deeper semantic information embedded in background knowledge, thus limiting the comprehensiveness and accuracy of the prediction models.

To address these issues, we propose MoF, a background-aware multi-source fusion financial trend forecasting mechanism. MoF integrates background knowledge regarding the impact of real-time events on stock price movements and utilizes a large language model to extract critical information from policy documents and stock commentaries, subsequently employing the MacBERT model to generate feature vectors for stock prediction. MoF leverages the comprehensive analysis and interpretive capabilities of large language models to capture deep semantic and emotional information from policy texts, thus providing richer background knowledge for stock trend prediction. By incorporating the influence of real-time events, this fusion mechanism effectively enhances the accuracy and interpretability of the predictions.

Our contributions are fivefold:

- The combined analytical understanding and information extraction capabilities of large-scale language models are utilized to summarize policy texts. Key information is extracted from policy texts using this approach, which effectively enhances the sensitivity of MoF to policy impacts.
- The pre-trained Chinese language model MacBERT is employed to extract feature vectors from policy summaries and stock review summaries, which are then fused with stock data. This fusion enriches data dimensions and fully considers the deep semantic and sentiment information present in various texts. It provides a more nuanced and accurate dimension of semantic information, addressing the limitations of traditional sentiment classification methods, which often fail to grasp the content and potential impact of the text.
- To account for the impact of historical stock movements on future trends, MoF incorporates a prior effect module. This module allows MoF to take historical stock movements into account when making predictions, thus enhancing its ability to learn from sequential data. Consequently, the accuracy and interpretability of the MoF are significantly improved.
- To the best of our knowledge, we are the first to propose a method that utilizes complex background knowledge along with LLM. Existing methods mainly focus on simple positive, negative, and neutral sentiment classification of text messages such as news texts and forum messages. However, these methods miss significant semantic information. Our approach is more adept at capturing the effects of policy changes and real-time events on stock prices, thereby offering a more comprehensive view. Additionally, we developed a straightforward UI that enables users to interact with the mechanism without prior knowledge of machine learning or expertise in large language models.
- In the experimental results section, when comparing the performance of the baseline model, our method may not reach the level of previous studies in some metrics. One reason previous research methods exhibit high prediction accuracy

is that they mainly forecast stock prices on day n using structural data, such as stock prices, trading volumes, and technical indicators from the preceding $n-1$ days, which can lead to some overstatement. MoF incorporates precise semantic analyses of background knowledge based on financial time series, which, to some extent, offers the correct research direction.

Paper Organization The rest of the paper is organized as follows. Section 2 presents the recently advanced background knowledge of our approach. Section 3 details the approach components of MoF. Section 4 reports our experimental results on our mechanism. Section 5 outlines the most related work. Section 6 concludes the paper with a future research discussion.

2 Background

In this section, we mainly demonstrate the knowledge of some recently advanced technologies.

2.1 Pre-trained Models

Pre-trained Models (PTM) (Han et al., 2021) refer to neural network models that have been previously trained on large-scale datasets. These models typically learn features from general tasks that can be transferred to specific tasks. The advantage of pre-trained models is that they can utilize the statistical information from large-scale data to learn universal language representations, thereby eliminating the need for expensive data labeling and model training for specific tasks. Once pre-training is completed, these models can be easily fine-tuned or transferred to meet the requirements of particular domains or tasks.

In recent years, due to the development of deep learning and large-scale data, pre-trained models have achieved great success in the field of Natural Language Processing (NLP) (Ruder et al., 2019). The emergence of Transformers has particularly revolutionized the landscape of pre-trained models. Transformer models achieve better contextual understanding through self-attention mechanisms, enabling them better to capture the global semantic information of sentences and documents (Vaswani et al., 2017). BERT (Devlin et al., 2018), GPT (Radford et al., 2018), and other Transformer-based models have been widely applied in NLP tasks. These large-scale PTMs can capture multiple disambiguations, lexical and syntactic structures, and factual knowledge from texts (Han et al., 2021). By fine-tuning these large-scale PTMs on extensive samples, their rich linguistic knowledge results in superior performance in downstream NLP tasks (Howard & Ruder, 2018; Liu et al., 2019).

2.2 Large Language Model (LLM)

A Large Language Model (LLM) is a deep learning model capable of understanding and generating natural language text. It is usually based on a neural network

architecture with a large number of parameters and a rich language representation (Vaswani et al., 2017; Radford et al., 2018).

A significant advantage of LLMs is their powerful generative capabilities. By learning large amounts of textual data, these models are able to generate high-quality, coherent text (Radford et al., 2018). Multi-tasking capability is another major advantage of big language models. While traditional NLP models usually need to be trained individually for each task, LLMs are able to achieve excellent performance on multiple tasks through pre-training and fine-tuning (Devlin et al., 2018; Ruder et al., 2019). This multi-tasking capability not only improves the efficiency of model application but also reduces development costs and time. Typical tasks include text generation, translation, Q&A systems, summary generation, sentiment analysis and text classification (Vaswani et al., 2017; Radford et al., 2018). Large Language Models can be quickly adapted to new tasks through a simple fine-tuning process, demonstrating flexibility and adaptability (Ruder et al., 2019).

In terms of language understanding, LLMs show deep analysis and comprehension capabilities. They can capture complex patterns and long-distance dependencies in language, providing a comprehensive understanding and processing of contextual information. This ability allows LLMs to excel in tasks, e.g., reading comprehension, information extraction, and dialogue systems (Vaswani et al., 2017). For example, the BERT model is able to understand the relationship between each word in a sentence and other words through a bidirectional encoder structure, thus providing more accurate semantic understanding (Devlin et al., 2018).

2.3 MacBERT

MacBERT is an improved version of the BERT-based model designed to improve the performance of Chinese language processing tasks. It replaces the original MLM task as MLM as Correction (Mac) task by masking words using words similar to them, reducing the gap between the pre-training and fine-tuning phases. This masking strategy proves to be very effective in downstream tasks. MacBERT is designed to provide a powerful and effective pre-trained language model to advance the field of Chinese natural language processing research (Cui et al., 2020).

MacBERT employs larger and more diverse Chinese text data for pre-training to improve its comprehension of the Chinese language. Different from traditional BERT models, MacBERT introduces innovations in its masking strategy. Instead of using the traditional [MASK] notation to mask the target word, it uses words similar to the target word for replacement. This similar word substitution strategy helps to retain semantic information better and improves the performance of the model in fine-tuning tasks. In addition, MacBERT employs whole-word masking and N-gram masking strategies to enhance the comprehension of long texts and improve the ability to capture contextual information of the model. For tasks similar to the NSP task, the Sentence Order Prediction (SOP) task is performed. These designs enable MacBERT to excel in handling various Chinese natural language processing tasks (Cui et al., 2020).

3 Approach

In this section, we provide a comprehensive overview of the methodologies underpinning our mechanism. The key components include: the integration of background knowledge, the deployment of large language models, the extraction of textual information features, and the training of predictive models for stock price fluctuations. We then detail the design of the Background-aware multi-source fusion financial trend forecasting mechanism, hereinafter referred to as MoF. The overall structure of MoF is shown in Fig. 1. The workflow of MoF can be described as a four-step process. First, MoF takes preprocessed stock price movement data as input, obtaining information, e.g., stock date (Date) and price change (Chg), and represents it as binary labels of 0 and 1. Second, MoF uses a sliding window to capture the price movements of the preceding $n-1$ days relative to the target day, representing them as binary labels and combining this with the subsequent price movement data to be predicted. Third, MoF further extracts features from textual information, e.g., policy information and stock comments, and inputs them along with the price movements of the preceding $n-1$ days into a time series forecasting model network. Finally, MoF forwards the results from the third step to a fully connected layer to predict stock price movements and output binary labels.

3.1 Mechanism Design Overview

Our forecasting mechanism is designed as a multi-source fusion model that combines the advantages of large language models with traditional financial data analysis. The overall goal is to leverage the language model to extract meaningful insights from textual data and integrate these insights with historical stock price data to improve the accuracy of forecasts.

Multi-Source Data Integration Our forecasting mechanism is conceptualized as a multi-source fusion model that integrates a plethora of data types. At the core of

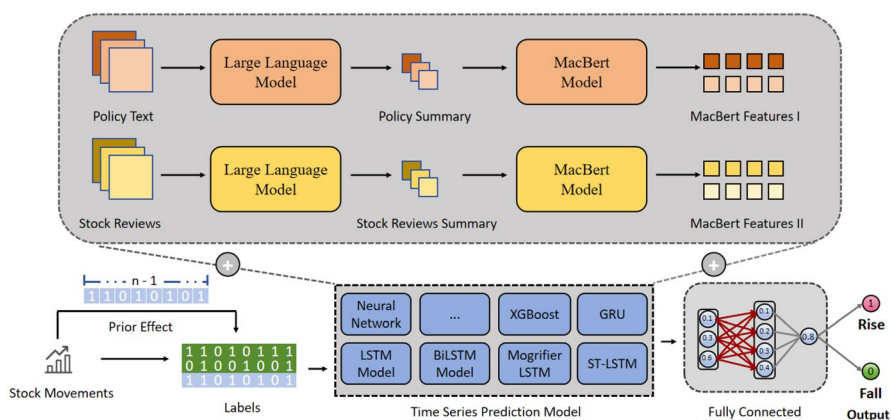


Fig. 1 Model architecture

this integration lies the amalgamation of the Large Language Model with traditional financial data analysis, thereby creating a comprehensive mechanism for stock price prediction.

Linguistic Model Component The linguistic component of our model focuses on extracting key information from textual data to create a summary using a large language model. Subsequently, the MacBERT model processes and refines this policy summary information into MacBERT Feature I.

Financial Data Processing Parallel to linguistic processing, the mechanism also ingests Stock Reviews, which are similarly processed to yield Stock Review Summaries. These summaries are then converted into a set of features known as MacBERT Features II, capturing the sentiment and trends present in the stock market discourse.

Fusion Mechanism The innovation of the mechanism lies in its fusion mechanism, where MacBERT Features I and MacBERT Features II are combined. This fusion facilitates a richer, multidimensional representation of the data, which is critical for accurate time series prediction of stock movements.

Time Series Prediction Model The integrated features are then fed into a Neural Network, which serves as the backbone of our time series prediction model. This network encompasses a variety of deep learning architectures, e.g., LSTM, BiLSTM, GRU, MogrifierLSTM, and ST-LSTM, each designed to capture different aspects of temporal dependencies within the stock price data.

Prior Effect Incorporation To account for the influence of past stock movements on future trends, the model incorporates a Prior Effect module. This module enables the mechanism to consider historical stock movements when making predictions, thus enhancing the ability of our model to learn from sequential data.

Output Layer Design The pivotal processing in our model takes place at the Fully Connected Output Layer, where final predictions are generated. This layer is tasked with translating the complex interactions of the integrated features and temporal dynamics into a binary output indicative of future stock movements.

Usability-Enhanced Interface To address the challenge of accessibility posed by the sophisticated architecture of the MoF mechanism, particularly for those without expertise in advanced machine learning, we have crafted an intuitive user interface as shown in Fig. 2 that simplifies the utilization process for users irrespective of their knowledge in machine learning.

This systematic design approach, with its multi-tiered architecture and sophisticated integration of linguistic and financial analytics, aims to push the boundaries of financial trend forecasting by providing a nuanced, data-driven perspective on stock price movements.

3.2 Data Collection and Preprocessing

Our approach begins with a well-designed data collection process, incorporating A-share market data, e.g., the Shanghai and Shenzhen indices. Additionally, we gathered information on policies released by the People's Bank of China (PBOC) during the same period and utilized a large language model to generate summaries of these policies, providing insights into the macroeconomic context. We also collected legally

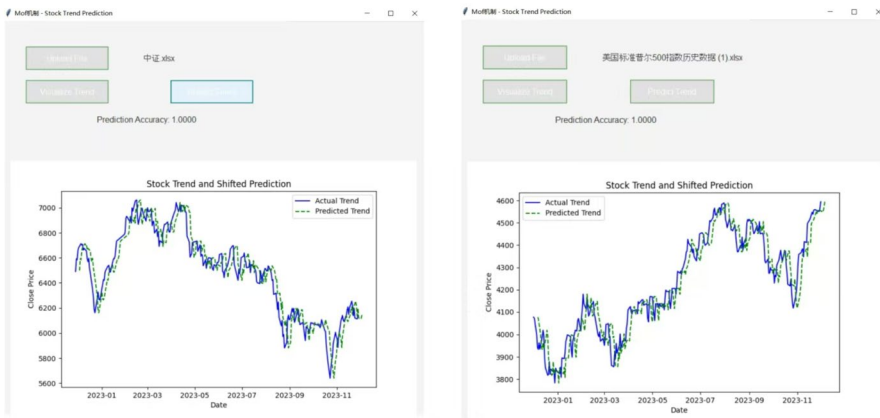


Fig. 2 MoF mechanism user-friendly UI interface

obtained stock reviews from the same period and processed them using a large language model for future use. Data preprocessing is a critical step to ensure data standardization and readiness for subsequent analysis.

Data Parsing Data parsing is a critical process in our mechanism, involving several key steps to ensure accurate analysis. Initially, dates from various sources are standardized to maintain consistency, aligning stock prices with relevant events and policies. Stock price movements are then converted into binary labels (e.g., ‘1’ for an increase, ‘0’ for a decrease) to simplify prediction. Policy documents are processed using advanced natural language processing to extract concise and relevant summaries. Rigorous data cleaning and normalization follow, ensuring the input data is of the highest quality. These steps collectively enhance the accuracy and reliability of our stock price prediction models.

Data Standardization To eliminate dimensional discrepancies across different datasets, we standardized stock price movement data using the following methods:

Min–max Normalization Scale the data to between 0 and 1, x represents each stock price data point that is being normalized to a range between 0 and 1, the formula is:

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)}, \quad (1)$$

Z-score normalization Normalized according to the mean (μ) and standard (σ) deviation of the data:

$$x_{std} = \frac{x - \mu}{\sigma}, \quad (2)$$

3.3 Textual Feature Extraction

Leveraging the model named MacBert, we process policy summaries to extract high-dimensional feature vectors. These vectors encapsulate deep semantic and sentiment information from the text, which is crucial for understanding the nuanced impact of policies on stock prices. Through the encoding process, MacBERT transforms policy summary text into distilled representations that capture the essence of the potential market impact of the policy. During textual feature extraction using the MacBERT model, we delve into its internal mechanisms to accurately capture deep semantics and sentiment in policy texts.

The Pre-trained MacBERT Model The MacBERT model learns diverse linguistic features during pre-training, laying a solid foundation for subsequent fine-tuning tasks (Cui et al., 2020). One of the pre-training tasks is the Masked Language Model (MLM), whose objective function can be represented as:

$$\mathcal{L}_{\text{MLM}} = - \sum_{i=1}^N \log p(t_i), \quad (3)$$

where N is the number of masked tokens, t_i is the index of the i -th token, and $p(t_i)$ is the predicted probability of the model for that token.

Advanced Masking Strategy and Model Interpretability A distinctive feature of MacBERT is the advanced masking strategy, which enhances the ability of our model to handle the intricacies of the Chinese language by substituting target words with semantically similar words during pre-training. The textual features extracted via MacBERT boost the predictive power and interpretability of our model, allowing stakeholders to gain insights into specific aspects of policy texts influencing predictions.

Semantic and Sentiment Information Capture The MacBERT model employs a bidirectional Transformer architecture, capturing relationships between words through the self-attention mechanism (Cui et al., 2020; Vaswani et al., 2017). The core of the self-attention mechanism is calculating the attention scores for each word with respect to all other words, as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (4)$$

where Q , K , V represent the Query, Key, and Value matrices, respectively, and d_k is the dimension of the keys.

In stock forecasting, the Query (Q) represents the feature to be predicted, e.g., historical prices and trading volumes of a particular stock, with our method specifically using closing prices. The Key (K) corresponds to the Query and identifies relevant information through characteristic vectors derived from policy documents, capturing critical economic and regulatory factors influencing stock prices. The Value (V) vector comprises the empirical data points used to generate the final prediction, including detailed historical performance data like price

movements and market trends, with our method specifically indicating the rise or fall of the stock (Vaswani et al., 2017).

The attention mechanism assigns weights to different parts of the input sequence:

$$\alpha_{ij} = \frac{\exp(\text{score}(h_{t-1}, F_j))}{\sum_k \exp(\text{score}(h_{t-1}, F_k))}, \quad (5)$$

this allows the model to focus on relevant features for the prediction task.

Positional Encoding To enable the model to understand the order of words in a sequence, positional encoding is added to the input embeddings in MacBERT, with the formulas:

$$\text{PE}_{(pos, 2i)} = \sin(pos/10000^{2i/d_{\text{model}}}), \quad (6)$$

$$\text{PE}_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}}), \quad (7)$$

where pos is the position of the word, i is the dimension index, and d_{model} is the dimension of the model.

Multi-Head Attention Mechanism MacBERT uses a multi-head attention mechanism to capture representations of information in multiple subspaces in parallel, with the formula:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O, \quad (8)$$

where h is the number of heads, head_i is the attention output of the i -th head, and W^O is the output weight matrix.

Layer Normalization and Residual Connections To improve the stability and efficiency of model training, MacBERT uses layer normalization and residual connections in each sub-layer:

$$\text{LayerNorm}(x + \text{Sublayer}(L(x))), \quad (9)$$

where $L(x)$ is the sub-layer (e.g., multi-head attention and feed-forward network), and LayerNorm is the layer normalization operation.

Feed-Forward Network Following the self-attention layer, MacBERT uses a feed-forward network to further process features, generally in the form of:

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2, \quad (10)$$

where W_1, W_2 are weight matrices, and b_1, b_2 are bias terms.

Feature Vector Formation: After processing by the encoder, each word is represented by an output vector h_i , which encapsulates contextual information. The output vector h_{start} of the first word (usually a special start token) is used as the representation of the entire sentence for further processing.

3.4 Model Architecture

The model architecture of our financial trend forecasting mechanism is a symphony of layers designed to integrate textual features with stock price time series data effectively. Here, we outline the role of each layer and its mathematical underpinning.

Input Layer The model receives as input the feature vectors from the MacBERT model and the historical stock price data, represented as a sequence $X = \{x_1, x_2, \dots, x_t\}$, where x_t is the stock price at time t .

Embedding Layer Textual features are embedded into a higher-dimensional space for richer representation:

$$E = W_e \cdot F + b_e, \quad (11)$$

where W_e is the weight matrix, F is the original feature vector, and b_e is the bias vector.

Convolutional Layer (For Text Data) We apply convolution to the embedded text features to extract local patterns:

$$C = \text{Conv}(E) = f(W_c \cdot E + b_c), \quad (12)$$

the function f denotes an activation like ReLU, W_c and b_c are the weights and biases of the convolutional layer.

Recurrent Layer The recurrent layer, e.g., LSTM (shown in Fig. 3), captures temporal dependencies with update rules:

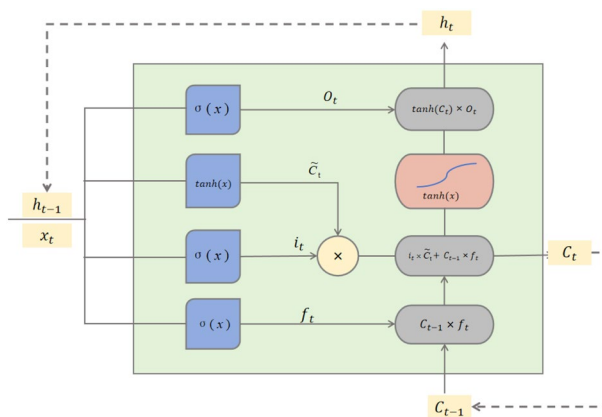


Fig. 3 LSTM

$$\begin{aligned}
i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i), \\
f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f), \\
\tilde{C}_t &= \tanh(W_C \cdot [h_{t-1}, x_t] + b_C), \\
C_t &= f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t, \\
o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o), \\
h_t &= o_t \cdot \tanh(C_t),
\end{aligned}$$

the i_t , f_t , and o_t represent the input, forget, and output gates, respectively, and $\circ$ denotes the Hadamard product.

Fusion Layer: The fusion layer combines textual and stock price data:

$$Z = \gamma \cdot O_t + (1 - \gamma) \cdot C, \quad (13)$$

the parameter γ balances the contributions from the output of the recurrent layer O_t and the convolutional layer C .

Fully Connected Layer The fused data is then passed through fully connected layers to perform classification:

$$Y = W_f \cdot Z + b_f, \quad (14)$$

where W_f and b_f are the weights and biases of the fully connected layer.

Output Layer A sigmoid function in the output layer provides the probability of an event, e.g., a stock price increase:

$$P = \sigma(Y), \quad (15)$$

Loss Function The binary cross-entropy loss is used to train the model to predict accurately:

$$\mathcal{L} = - \sum_t y_t \log(P_t) + (1 - y_t) \log(1 - P_t), \quad (16)$$

the y_t represents the actual target values and P_t the predicted probabilities.

3.5 Details for Model Training and Optimization

The following sections detail the training and optimization methods for our model, which refers to a time series prediction model embedded in the MoF mechanism.

Data Partitioning and Sequence Formulation Our methodology begins with a strategic division of the dataset into training and testing subsets, adhering to an 80-20 ratio. This partitioning is pivotal for ensuring the robustness and generalizability of our model's predictive capabilities. The formula for calculating the size of the training set based on the total data size T is:

$$\text{Training Set Size} = 0.8 \times T, \quad (17)$$

Model Training and Optimization The training regimen utilizes the Adam optimizer, an adaptive learning rate optimization algorithm known for its efficiency in handling

sparse gradients and noisy updates. We employ the binary cross-entropy loss function, which is particularly suited for binary classification tasks such as ours. The binary cross-entropy loss $\mathcal{L}_{\text{BCE}}$ is given by:

$$\mathcal{L}_{\text{BCE}} = - \left(\sum_{c=1}^C y_{o_c} \log(p_c) + (1 - y_{o_c}) \log(1 - p_c) \right), \quad (18)$$

where C is the number of classes, y_{o_c} is the binary indicator (0 or 1) if class c is the correct classification for the observation, and p_c is the predicted probability that the observation is of class c .

Evaluation Metrics and Performance Benchmarking The efficacy of our model is rigorously evaluated using accuracy as the primary metric. Accuracy provides a quantitative measure of the ability to correctly predict stock price movements, offering a straightforward assessment of its predictive prowess. The formula for accuracy A is:

$$A = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}, \quad (19)$$

To further validate our model's superiority, we conduct a comparative analysis against a spectrum of established models, including but not limited to Neural Network, BiLSTM, LSTM, XGBoost, GRU, STLSTM, and MogriflerLSTM. This comparative analysis serves as a benchmark, highlighting our model's enhanced prediction accuracy and interpretability.

Comparative Analysis and Model Interpretability The comparative analysis not only benchmarks the performance of our model but also offers insights into its interpretability. By juxtaposing our model's predictions with those of other models, we can deduce the extent to which the integration of linguistic features and temporal stock data contributes to the model's predictive power. We also consider other metrics, e.g., the F1 score, which combines the precision P and recall R into a single measure:

$$F1 = 2 \times \frac{P \times R}{P + R}, \quad (20)$$

This metric is particularly useful when the class distribution is imbalanced.

This integrated approach to experimental design and model evaluation ensures a comprehensive assessment of our forecasting mechanism, positioning it as a robust and reliable tool for stock market analysis.

4 Evaluation

In this section, we provide a detailed evaluation of MoF. We will first introduce our hardware setup. Next, we describe the dataset that we collected and processed. Following this, we focus on comparing the performance of various time series prediction models integrated into the mechanism, encompassing both traditional models and newer variants.

Additionally, we present detailed case studies to facilitate thorough comparisons, further exploring and elucidating the role and potential of our mechanism.

4.1 Experimental Environment

The following describes the hardware devices and data set information we used in the evaluation process. Hardware information includes computer configuration, processor, memory and other key information. Data set information includes the data set we use, including data sources, data types, data preprocessing methods, etc.

4.1.1 Experimental Configuration

Our experimental computer was equipped with a ninth-generation Intel Core i7 processor and an NVIDIA GeForce GTX 1660 Ti stand-alone graphics card with 8GB of RAM. Our experimental software setup includes PyTorch 2.0.0 and CUDA 11.7, running on a Windows 10 operating system. For regular and fully connected layers, we employed the Adam optimizer with a learning rate of $1e-3$ and a batch size of 32.

4.1.2 Dataset

We mainly collected A-share market data, including the Shanghai Composite Index, Shenzhen Composite Index and China Securities Index. Regarding policy information, we extracted policy summaries from the People's Bank of China (PBOC) using a large language model.

Stock Data We collected A-share market data from reliable financial information sources, including indices, e.g., the Shanghai Securities Composite Index (SSEC), the Shenzhen Composite Index (SZI) and the China Securities Index (CSI), the Us Standard & Poor's 500 index (US S&P). These four indexes are important tools for investors to understand and analyze the domestic and foreign stock market, and they each have different calculation methods and component compositions, which can show the dynamics and trends of the domestic and foreign stock market from various angles. To facilitate visualization of the data sets, we drew their heat maps, as shown in Figs. 4, 5, 6a and b.

The data regarding stock price movements is derived from these three indexes, covering the period from December 2022 to December 2023, with a daily frequency. Through our processing, we organize the data into a tabular format that includes data, change (Chg), opening price, closing price, trading volume, and other indicators. A data preprocessing step is applied to the stock price change information. It involves parsing the dates, converting the stock price change (Chg) into floating-point numbers, and categorizing them into binary labels (1 for increase, 0 for decrease).

Policy Information We obtained policy summaries from the People's Bank of China (PBOC) through a legitimate web crawler or existing database. Text features are extracted from these summaries, where non-string values are replaced with

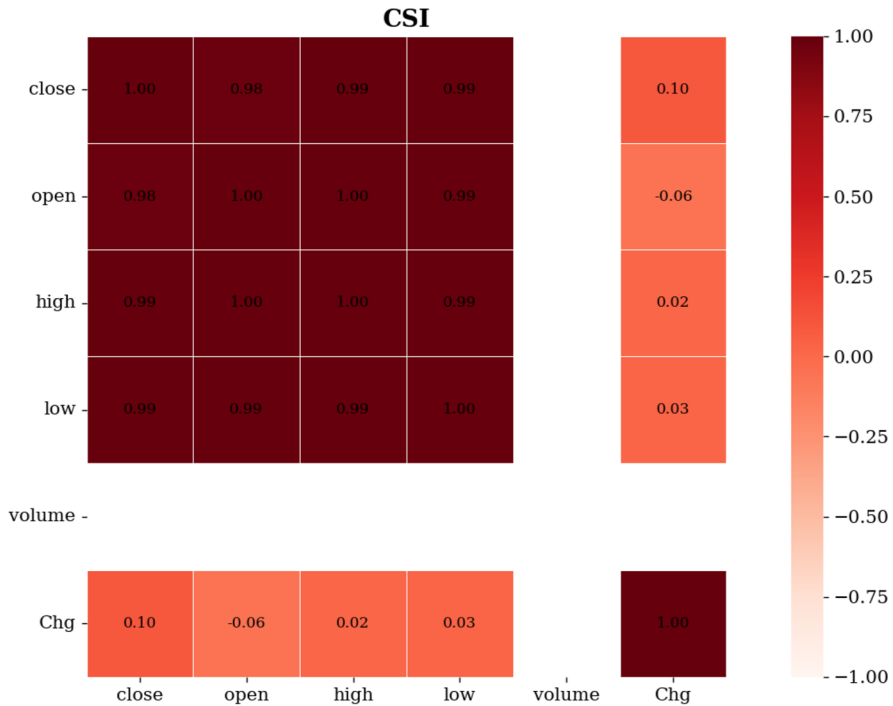


Fig. 4 CSI heat map

NaN. The textual features are represented as multiple high-dimensional matrices. To ensure consistency in subsequent prediction processes, we standardized these text features by reducing the dimensionality of the high-dimensional matrices to one dimension. We then filled or truncated them to a fixed length to ensure uniformity across the dataset, as shown in Table 1.

4.1.3 Evaluation Index

We used the four metrics most commonly used in machine learning-related research, i.e., accuracy, F1, precision value, and recall rate.

Accuracy Accuracy is the proportion of the number of correctly classified samples to the total number of samples, and it is an intuitive indicator. Accuracy is calculated as follows:

$$\text{Accuracy} = \frac{\text{Correct Sample}}{\text{Sample Count}} \times 100\%, \quad (21)$$

Precision Accuracy is the proportion of positive categories that the model predicts (such as a rising stock) are actually positive, and a high accuracy means that the model rarely misclassifies negative categories into positive ones. Precision is calculated as follows:

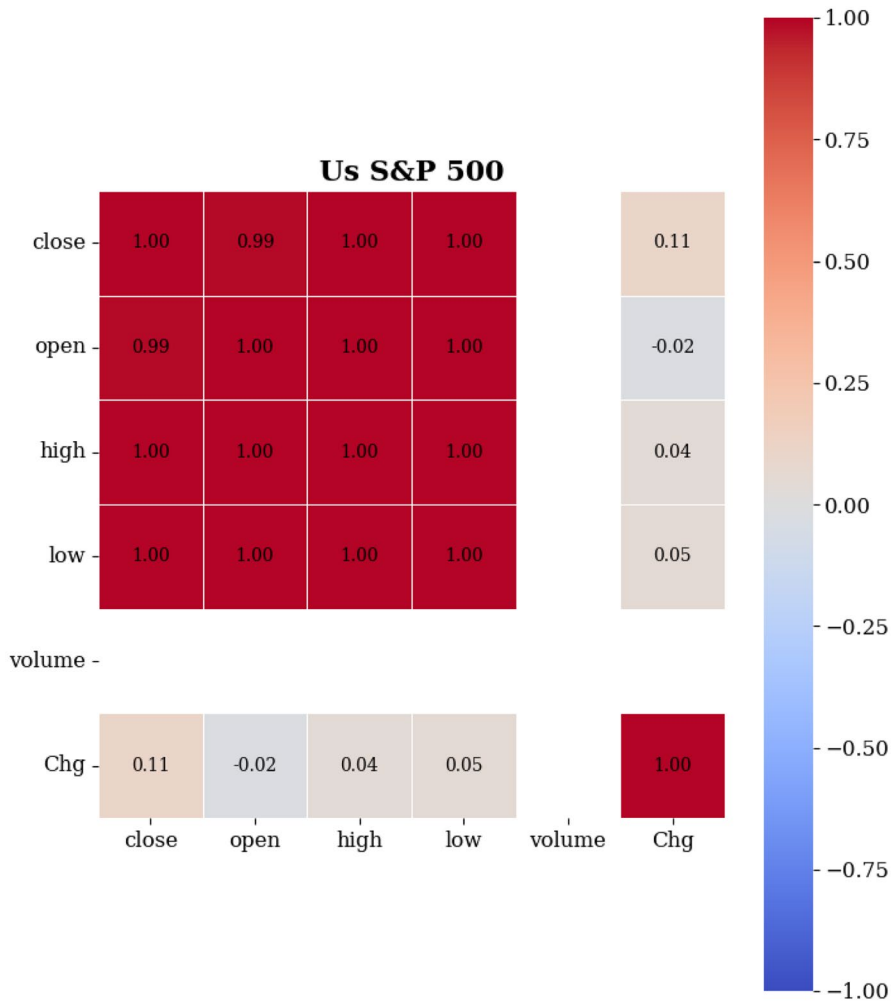
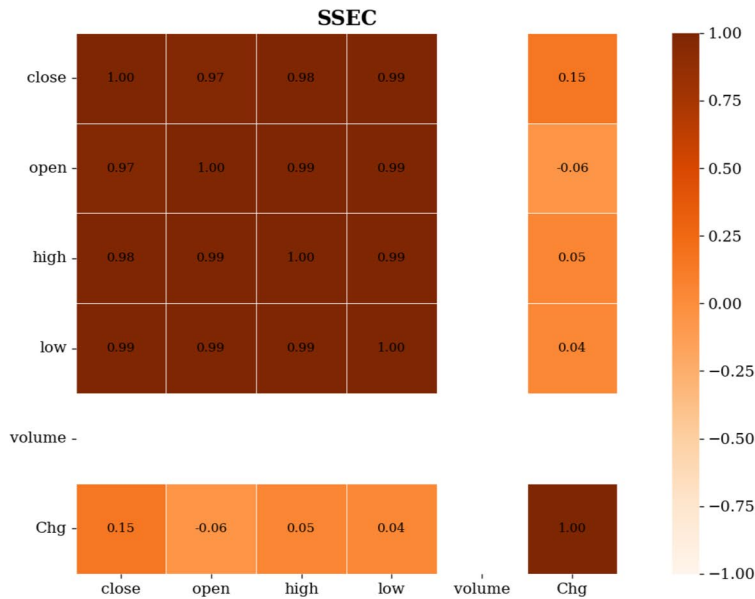


Fig. 5 US standard & poor's 500 index heat map

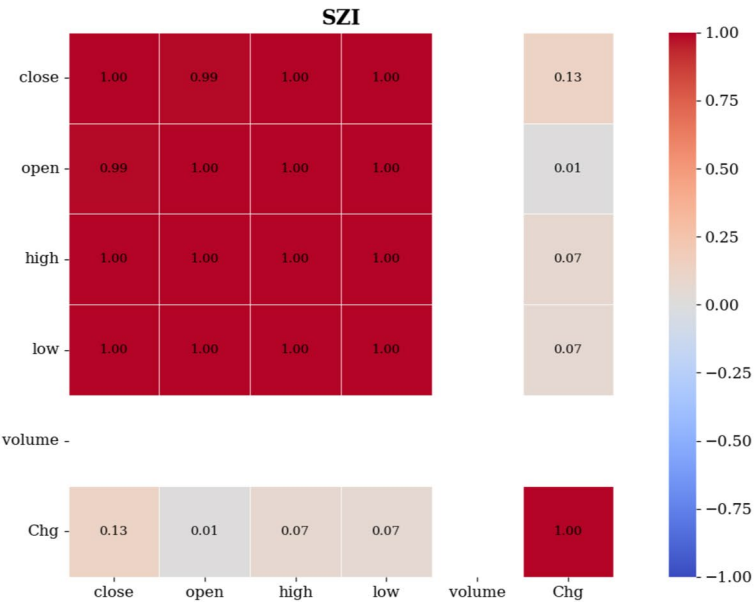
$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \times 100\%, \quad (22)$$

Recall The recall rate, also known as the true rate or sensitivity, is the proportion of samples that are actually in the positive category that the model correctly predicts to be in the positive category. The high recall rate means that the model is able to capture most of the positive class samples. Precision is calculated as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \times 100\%, \quad (23)$$



(a) SSEC



(b) SZI

Fig. 6 Stock data heat map

Table 1 Examples of policy text information and predicted values

Date	Policy text (English version)	Value
2023/01/03	State Administration of Taxation and all relevant tax bureaus: In 2023, the State Administration of Taxation will continue the “Spring Breeze Action” for the 10th year, ... Tax authorities should enhance leadership and innovation to boost taxpayer satisfaction and support modernization	1
2023/11/02	To financial institutions: The Financial Regulatory Bureau has issued the “Capital Management Measures for Commercial Banks,” ... Compliance plans and reporting on progress are also mandated	0
2023/10/30	To guide state-owned insurance companies, the Ministry of Finance has issued new guidelines. ... The guidelines take effect immediately for 2023 evaluations	1
2023/09/27	To support rural finance, corporate income tax policies are extended: interest from small loans to farmers and insurance premiums for agriculture are taxed at 90%. ... to apply the policy	1

The table delineates a case study of policy text information in English, i.e., each date is associated with directives issued by either the Securities Regulatory Commission (SRC) or the State Council (SC). A value of ‘1’ signifies an anticipated appreciation in stock prices, whereas a value of ‘0’ denotes an expected depreciation

F1 Score The F1 score is a harmonic average of accuracy and recall, and it attempts to consider both metrics at the same time. F1 scores are balanced between accuracy and recall and are an indicator that takes both into account. F1 is calculated as follows:

$$\text{F1 Score} = 2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}}. \quad (24)$$

4.2 Time Series Prediction Model

We embed eight models that can be used for time series prediction into the mechanism, fully compare the performance of each model, and analyze the advantages of each model:

4.2.1 Neural Network

In time series prediction, neural networks are renowned for their robust nonlinear modeling capabilities, automatic feature extraction, ability to process multi-variable inputs, capture sequence dependencies, and exhibit strong generalization ability and

scalability. Despite their higher computational demands and data volume requirements, along with risks of overfitting and interpretability challenges, neural networks, when properly trained, are effective for time series prediction tasks, providing accurate and reliable results (Zhang et al., 1998).

4.2.2 Long Short Term Memory Network

LSTM (Long Short-Term Memory Network) stands out as a specialized type of recurrent neural network that addresses the issues of gradient vanishing or exploding in traditional RNNs through gating mechanisms. It is particularly well-suited for time series prediction due to its capability to capture long-term dependencies, autonomously learn and extract critical features, handle nonlinear relationships within sequence data, and demonstrate strong generalization ability. Moreover, the capacity of LSTM to handle sequences of varying lengths enhances its flexibility and efficacy in time series forecasting (Hochreiter & Schmidhuber, 1997).

4.2.3 Bidirectional Long Short Term Memory Network

BiLSTM (Bidirectional Long Short-Term Memory Network) enhances its capability to capture long-term dependencies and intricate patterns in time series prediction by considering both preceding and succeeding information in the sequence. This approach significantly improves prediction accuracy and robustness, albeit at the cost of increased computational resources and data requirements (Graves & Schmidhuber, 2005).

4.2.4 eXtreme Gradient Boosting

XGBoost (eXtreme Gradient Boosting) is renowned in time series prediction for its efficient gradient boosting algorithm, exceptional ability to handle nonlinearity and large feature sets, and effective regularization techniques to prevent overfitting. It supports custom functions and parallel computation, thereby enhancing model performance and accelerating training speed (Chen & Guestrin, 2016).

4.2.5 Mogrifler Long Short Term Memory Network

MogriflerLSTM (Mogrifler Long Short-Term Memory Network) is a variant derived from LSTM that augments the expressive capacity of the LSTM network by introducing additional parameters. This adaptation enables MogriflerLSTM to effectively capture more complex patterns inherent in time series data with highly intricate and nonlinear characteristics, thereby improving prediction accuracy (Melis et al., 2019).

4.2.6 Gated Recurrent Unit

Gated Recurrent Unit (GRU) is a simplified version of LSTM that controls the flow of information through a gating mechanism. It captures long-term dependencies in time series data while having fewer parameters and a simpler structure, which

generally makes GRUs faster to train and reduces the risk of overfitting. GRU is particularly suitable for processing time series data with fewer long-term dependencies and can be comparable to LSTM in some cases, but it is more efficient in terms of the number of parameters and computational complexity (Cho et al., 2014).

4.2.7 Spatio-Temporal Long Short-Term Memory

ST-LSTM (Spatio-Temporal Long Short-Term Memory) is a variant of LSTM designed for spatio-temporal data. By integrating spatial and temporal features, it improves the ability to capture complex dynamics, making it especially suitable for multi-dimensional time series prediction tasks, although it requires more computational resources (Tang et al., 2019).

4.2.8 Swin Transformer Long Short-Term Memory

SwinLSTM (Swin Transformer Long Short-Term Memory) is a neural network unit that combines the self-attention mechanism of the Swin Transformer with the recurrent structure of LSTM. It captures global spatial dependencies through the self-attention mechanism and maintains long-term temporal dependencies through the gated unit of LSTM. This combination enhances the expressiveness and prediction accuracy of the model in time series prediction, making it especially suitable for processing sequence data with complex temporal and spatial characteristics (Du et al., n.d.).

4.3 Experimental Results

We first compared the accuracy of six models (LSTM, BiLSTM, MogrifierLSTM, GRU, ST-LSTM, SwinLSTM) embedded in MoF for predicting the rise and fall of stocks in the China Securities Index dataset, as shown in Table 2. The results indicate that, in our mechanism, due to the inclusion of policy information and other influential characteristics, the accuracy of predicting stock movements is generally favourable.

Table 2 Experimental results on China securities index

Models	100ep		200ep		400ep	
	Acc	Pre	Acc	Pre	Acc	Pre
LSTM	0.7333	0.7501	0.6334	0.6667	0.6667	0.6875
BiLSTM	0.7241	0.7692	0.6896	0.7143	0.6897	0.7142
MogrifierLSTM	0.6897	0.6667	0.7241	0.7692	0.5862	0.6001
GRU	0.5517	0.5385	0.5517	0.5556	0.6897	0.7143
ST-LSTM	0.6552	0.6471	0.7931	0.9091	0.7586	0.8333
SwinLSTM	0.7667	0.8001	0.6667	0.7501	0.6666	0.7501

- *Multi-source data fusion enhances the prediction ability of the model* The MoF constructs a multi-dimensional data set by integrating stock price data, policy text summaries, and stock commentary summaries. The tabular data shows that the accuracy rate of ST-LSTM on the 400-epoch window reaches 0.7931, with a precision rate of 0.9091. This indicates that multi-source data fusion significantly improves the prediction accuracy of our model. This integration not only increases the richness of the data but also allows the models to capture more complex market dynamics and policy implications.
- *The model structure design improves the depth and breadth of time series prediction* The MoF employs deep learning architectures such as LSTM and its variants, which are particularly well-suited for working with time series data and capturing long-term dependencies. The high accuracy of ST-LSTM and Swin-LSTM in the table shows that the time series prediction layer of the model can effectively learn from stock price data and predict future trends. Additionally, the fully connected output layer of the model can translate complex temporal dynamics and text features into predictions, further enhancing the depth and breadth of the predictions of the model.
- *Optimization strategies ensure the efficiency and robustness of model training* The Adam optimizer and the binary cross-entropy loss function mentioned in this paper provide an effective training strategy for the model. The tabular data shows that the accuracy and precision rates of all models are generally higher across different time windows, reflecting the effectiveness of the optimization strategy. Additionally, hyperparameter tuning further ensures the robustness of the model during training and helps avoid overfitting.
- *The interpretability of the model provides transparency to investment decisions* The MoF model not only performs well in prediction accuracy but also enhances interpretability by using the text features extracted by the MacBERT model. This allows investors and analysts to understand better how the model makes predictions based on policy texts and market sentiment. The differences in performance among the various models in the table can help researchers identify which features are most critical to the forecast, thereby improving transparency and trust in investment decisions.

We compared five time series prediction models embedded in the MoF mechanism: LSTM, BiLSTM, MogriflerLSTM, ST-LSTM, and SwinLSTM. We conducted stock rise and fall forecasting training on two datasets, the Shenzhen Index and the Shanghai Index. The training was conducted for 100, 200, and 400 rounds, respectively, and the precision for each round corresponding to each dataset was recorded, as shown in Table 3. The results indicate that due to the inclusion of influential factors, e.g., policy information, the precision for these two datasets is quite high in our mechanism.

Focusing on the data and models in Table 3, the results highlight the predictive power of the MoF mechanism under different time windows and market conditions. The ST-LSTM model exhibits the highest precision of 0.9091 in a 400-epoch time window, demonstrating its strong ability to capture long-term dependencies. The MogriflerLSTM model shows stability in the 200-epoch and 400-epoch windows

Table 3 Precision comparison of Shenzhen composite index and Shanghai composite index predictions

Models	100ep		200ep		400ep	
	SZI	SSEC	SZI	SSEC	SZI	SSEC
LSTM	0.9999	0.9167	0.8333	0.9167	0.8889	0.8571
BiLSTM	–	–	0.9999	0.9999	0.9999	0.9167
MogrifierLSTM	0.9000	0.9999	0.8462	0.8667	0.8182	0.9999
ST-LSTM	0.9999	–	0.9090	0.9999	0.9091	0.9999
SwinLSTM	–	–	0.9999	0.8000	0.8000	0.7500

with precisions of 0.8462 and 0.8182, respectively, supporting the robustness of the mechanism. The LSTM model achieved a perfect precision of 0.9999 in the 100-epoch window and maintained high precision in the longer windows, showing its ability to adapt to market changes quickly. The BiLSTM model exhibits a high precision of 0.9999 in the 400-epoch window, highlighting its advantages in long-term forecasting. Additionally, the high precision of the ST-LSTM model, 0.9090 and 0.9091, on different market indices (SZI and SSEC) proves that the MoF mechanism can adapt to the characteristics of different market indices and provide customized forecasting services. Together, these findings highlight the advantages of the MoF mechanism in terms of precision, robustness, adaptability, and applicability, providing investors with a reliable stock market prediction tool.

The following conclusions can be drawn:

- *Effectiveness of Multi-Source Data Fusion is Optimistic* The model performance presented in Table 3 indicates that the fusion of multi-source data, e.g., stock price data, policy text summaries, and stock review summaries, can significantly improve prediction accuracy. This supports the effectiveness of the multi-source data integration approach in the design of the MoF mechanism and emphasizes the importance of considering various information sources in financial trend forecasting.
- *The Strong Capability of MacBERT for Text Feature Extraction* The MoF mechanism utilizes text features extracted by the MacBERT model, which plays a crucial role in enhancing prediction accuracy. The high accuracy results in Table 3 further confirm the ability of MacBERT to capture deep semantic and sentiment information from policy texts and market sentiment.
- *High Applicability of Time Series Prediction Models in MoF* The prediction results for different time windows in Table 3 demonstrate the applicability and effectiveness of LSTM and its variants (e.g., ST-LSTM) in time series prediction tasks. This supports the decision of the MoF mechanism to adopt these deep learning architectures for its time series prediction layer.
- *Positive Impact of Optimization Strategies on Model Performance* The MoF mechanism employs the Adam optimizer and binary cross-entropy loss function, combined with hyperparameter tuning, to ensure efficient and robust model training. The high accuracy performance of different models across various time

windows in Table 3 demonstrates the success of these optimization strategies in preventing overfitting and improving model generalization.

- *Generalization Ability of the Models* The data in Table 3 indicate that, despite fluctuations in performance across different time windows, most models maintain high precision consistently. This reflects the strong generalization ability of MoF under varying market conditions, providing investors with stable prediction results.
- *MoF has adaptability to overseas data* Tables 2 and 3 are all the predictions of Chinese stock market data. In order to observe the adaptability of MoF mechanism to overseas data, we added the historical data of the US Poor's 500 to make predictions, as shown in Table 4. The accuracy rate is above 70%, and the accuracy rate of XGBoost model reaches 0.8333. It is further indicated that MoF is applicable to both domestic and foreign data.

We also compared the training losses of six time series prediction models embedded in the MoF mechanism using the CSI dataset. These models include LSTM, BiLSTM, GRU, MogrifierLSTM, ST-LSTM, and SwinLSTM. We recorded the loss values for each training epoch, plotted them as scatter plots, and conducted a comparative analysis. Combining these findings with the innovations of our MoF mechanism, we derived the following insights:

- *The advantages of multi-source data fusion are evident* The MoF mechanism constructs a multi-dimensional dataset by integrating stock price data, policy text summaries, and stock commentary summaries. The loss graph in Fig. 4 shows that the accuracy of the model in predicting stock fluctuations is generally optimistic due to the inclusion of influential features, e.g., policy information. This indicates that multi-source data fusion significantly enhances the accuracy of the model in predicting stock price changes.
- *The interpretability of our mechanism improves the training effectiveness for predicting stock movements* The MoF mechanism not only achieves high prediction accuracy but also enhances model interpretability by leveraging text features extracted by the MacBERT model. This transparency allows stakeholders to understand better how the model uses policy texts and market sentiment in mak-

Table 4 Accuracy comparison of US S & P 500 index historical data predictions

Models	200 epochs	400 epochs
LSTM	0.7241	0.7242
BILSTM	0.7242	0.7586
XGBoost	—	0.8333
Mogrifier LSTM	0.7200	0.6900
GRU	0.7242	0.7586
ST-LSTM	0.7241	0.7241

ing predictions, thereby improving the effectiveness of training for stock movement predictions.

- *The MoF mechanism demonstrates strong generalization capabilities* As depicted in Fig. 7, the loss for all models gradually decreases during the training process, indicating that the models effectively learn from the training data and enhance their predictive abilities. Moreover, the consistent reduction in loss demonstrates their robust generalization capabilities, enabling accurate predictions of unseen data.

4.4 Ablation Study

We conducted an ablation study on the two embedding models within this mechanism, examining their performance with and without the “Prior Effect.” The “Prior Effect” represents the rise and fall of the stock price in the previous N-1 days, encoded as a one-dimensional matrix consisting of 1 s and 0 s, where 1 indicates an

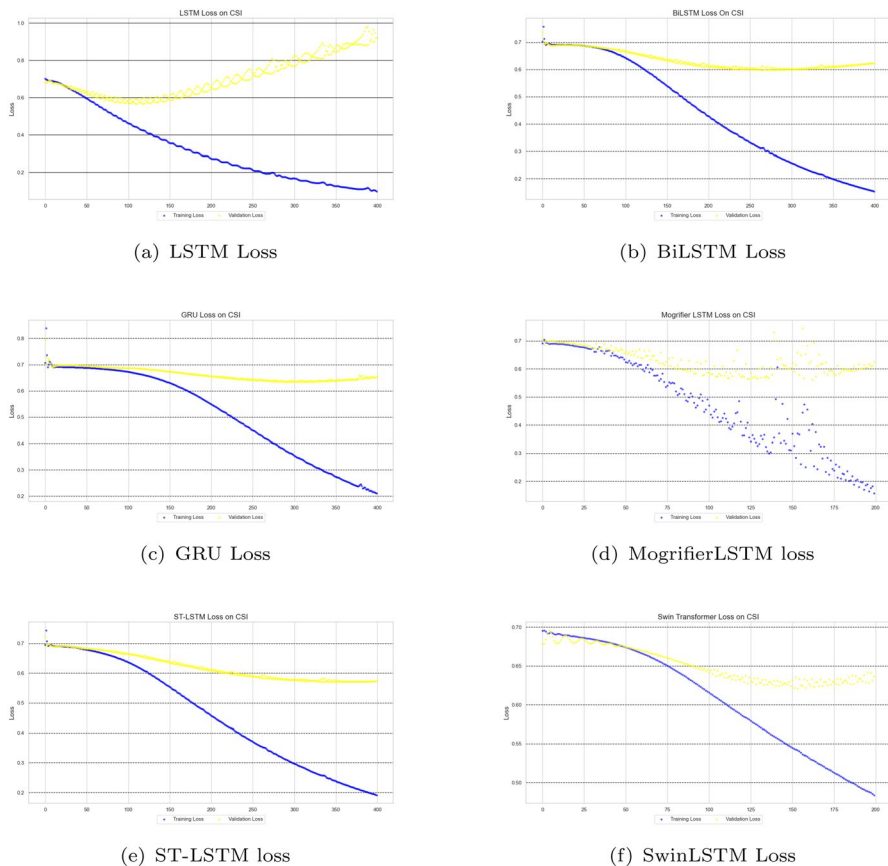


Fig. 7 Comparison of loss on CSI

increase, and 0 indicates a decline. The results demonstrate that the “Prior Effect” positively influences the prediction accuracy of stock rise and fall, as shown in Table 5.

Previous effects have a positive impact on the predictive power of the model. The ablation study reveals that the “Prior Effect” enhances the prediction of stock fluctuations. Specifically, without considering the “Prior Effect,” the Neural Network achieved an F1 score of 0.6177 and a recall rate of 0.5055. With the inclusion of the “Prior Effect,” the F1 score improved to 0.7428, and the recall rate increased to 0.8125. Similarly, for the LSTM model, the F1 score increased from 0.6451 with a recall of 0.6251 to 0.7692 with a recall of 0.6667. This underscores that incorporating the historical rise and fall of stocks over the past N-1 days significantly enhances the accuracy of the model in predicting future stock trends.

The explanatory power of the model is enhanced by the features extracted by the MacBERT model The MoF mechanism utilizes text features extracted by the MacBERT model, enhancing both prediction accuracy and model interpretability. This enables stakeholders to gain deeper insights into how predictions are derived from policy texts and market sentiment. In ablation studies comparing model performance with or without the “Prior Effect,” we can further elucidate the reasoning behind model predictions, thereby enhancing transparency in investment decisions.

Optimization strategies ensure the efficiency and robustness of model training The Adam optimizer and the binary cross entropy loss function mentioned in this paper provide an effective training strategy for the model. Data from the ablation study showed that the model maintained high accuracy and accuracy rates over different time windows with or without Prior Effect, reflecting the effectiveness of the optimization strategy. In addition, hyperparameter tuning further ensures robustness during model training and helps to avoid overfitting.

5 Related Works

The research on stock price prediction can be divided into three categories, i.e., traditional methods, machine learning-based methods, and deep learning-based methods.

Machine Learning-based Methods Dey et al. (2016) used the XGBoost algorithm to design an effective model to predict stock trends using technical indicators as features. Di Persio and Honchar (2017) compare three different RNN architectures, i.e., basic RNN, LSTM, and gated recursive unit (GRU), to assess which RNN performs better in predicting stock price movements of Google. Leung et al.

Table 5 Ablation study

Method	Without prior effect		With prior effect	
	F1	Recall	F1	Recall
Neural Network	0.6177	0.5055	0.7428	0.8125
LSTM	0.6451	0.6251	0.7692	0.6667

(2014) connect collaborating companies in the information technology sector in a graph structure and use an SSVM to predict positive or negative movement in their stock prices. Xiao et al. (2020) propose the cumulative autoregressive moving average method combined with the least squares support vector machine synthetic model (ARI-MA-LS-SVM) for basic stock market forecasting. Secondly, the data processing of predictive indicators is firstly carried out using cumulative autoregressive moving averages. Then, a least squares support vector machine using a simple indicator system is used to predict stock price fluctuations. Kohara et al. (1997) used a priority knowledge to extract event information and economic indicators, combined with a neural network model for multivariate stock price prediction, and the experimental results showed the high prediction accuracy and effectiveness of the method. De Fortuny et al. (2014) evaluated the applicability of different performance metrics by designing stock price prediction models based on text mining techniques, and explored how to evaluate, validate, and improve these models in real-world applications with the aim of improving the accuracy of stock price prediction. Soni et al. (2022) explore different techniques used to predict stock prices, from traditional machine learning and deep learning methods to neural networks and graph-based approaches. Vijh et al. (2020) used artificial neural networks and random forest techniques to predict the next day's closing prices of five companies belonging to different industries. Zhang et al. (2018) proposed a stock price trend prediction system that uses an unsupervised heuristic algorithm to predict stock price movements within a predetermined prediction time frame.

These studies primarily utilize machine learning methods to predict stock trends based on price time series data or technical indicators. While effective, they often lack the incorporation of background knowledge and deep semantic information from text data, limiting their predictive capabilities.

Deep Learning-based Methods Yang et al. (2017) introduced a method that integrates deep neural networks to model and forecast Chinese stock market indices, using recent indices as inputs. Lu et al. (2021) proposed a CNN-BiLSTM-AM approach to predict the next day's stock closing price. The technique consists of an Attention Mechanism (AM), a Bidirectional Long Short Term Memory (BiLSTM), and Convolutional Neural Network (CNN) for extracting features from the input data. Islam and Nguyen (2020) conducted a comparative study on stock price prediction using three different methods: autoregressive integrated moving average, artificial neural network, and stochastic process-geometric Brownian motion. Yu and Yan (2020) designed a deep neural network prediction model to forecast stock prices using the Phase Space Reconstruction (PSR) method and LSTM. Selvin et al. (2017) predict the stock prices of businesses listed on the NSE using three different deep learning architectures and compare their performance. Jin et al. (2020) propose a stock market prediction method that incorporates an attention mechanism into Long Short-Term Memory (LSTM) networks, applies Empirical Mode Decomposition (EMD) to break down time series, and fully accounts for investor sentiment. Based on the Long Short-Term Memory (LSTM) deep learning algorithm, Kyoung-Sook and Hongjoong (2019) presented a method for forecasting stock market indices and their volatility. Hossain et al. (2018) proposed a hybrid deep learning model that combines Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU).

They assessed the model's prediction ability using metrics like Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAP) by training and testing it on historical data from the S&P 500 index. Ghosh et al. (2019) provide a framework for analysing and projecting a company's future growth using the LSTM (Long Short-Term Memory) model and the firm's net growth calculation algorithm. Jing et al. (2021) proposed a hybrid model that utilises a convolutional neural network model to classify investor sentiment and applies a long short-term memory (LSTM) neural network approach to analyse the results of stock market technical indicators and sentiment analysis for stock price prediction. Li et al. (2020) conducted a comprehensive study on the predictability of investor sentiment, which was measured by extracting online user-generated content (UGC) expectations. The results of the study indicate that advanced models like LSTM can provide more accurate predictions of investor sentiment only when the model inputs contain predictive information. Swathi et al. (2022) proposed a Teaching and Learning Based Optimisation (TLBO) model and Long Short-Term Memory (LSTM) based sentiment analysis for stock price prediction using Twitter data.

These studies primarily utilize deep learning methods to predict stock trends based on price time series data or technical indicators without considering the contextual knowledge associated with stock price data. Even some approaches incorporate background knowledge through sentiment analysis of textual data. These approaches typically classify news text into positive, negative, and neutral categories and then use this information to predict stock prices. This method fails to fully leverage the deep semantic data information embedded in the text, limiting the potential accuracy of stock price predictions.

6 Conclusions and Future Research

This paper introduces a background-aware multi-source fusion financial trend forecasting mechanism (MoF), which integrates large-scale language models and time series analysis techniques to enhance the accuracy and interpretability of stock price predictions significantly. By fusing stock price data, policy text summaries, and stock commentary summaries, MoF constructs a multidimensional dataset that enables models to grasp market dynamics and policy implications comprehensively. MoF utilizes MacBERT, a pre-trained language model optimized for Chinese, to extract key information from policy texts and generate feature vectors that enrich model inputs. Employing the deep learning architecture of LSTM and its variants, MoF efficiently processes time series data and learns long-term dependencies to forecast future stock trends.

The optimization strategy, featuring the Adam optimizer and binary cross-entropy loss function, coupled with hyperparameter tuning, ensures efficient and robust model training while mitigating overfitting risks. Moreover, improvements in model interpretability allow stakeholders to understand the reasoning behind the predictions clearly. The ablation study further underscores the significance of the "Prior Effect," which considers the rise or fall of stock price over the previous N-1 days in enhancing forecast accuracy. Experimental results demonstrate that the MoF

mechanism performs well across various evaluation metrics and introduces significant innovations, providing a robust new tool for stock market prediction and investment decision-making.

While MacBERT has proven effective for extracting features from policy texts, certain aspects of the model's performance still offer room for improvement. Currently, the model relies on daily data for stock price predictions, which may not fully capture short-term fluctuations in the stock market. Future work could explore the use of higher-frequency data, such as minute-level or hourly stock data, combined with real-time information, to enhance the model's responsiveness to short-term market dynamics. Additionally, incorporating more advanced natural language processing techniques or integrating multi-modal data, such as macroeconomic and financial indicators, could further enrich the model's understanding of market trends and improve predictive accuracy.

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Data Availability The data are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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